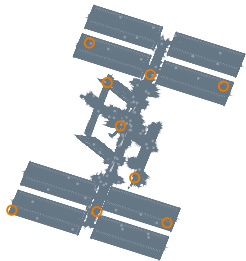


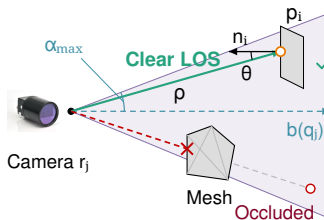
Viability-preserving rollout ADP for orbital inspection

a Mission inputs



○ Required targets K

b Camera and LOS



Range · FOV · incidence · clear LOS

Valid targets → G_j

$s = (j, m, b, h)$

c HCW transfer → SOOA graph

1 Solve HCW

Rest-to-rest · 90 s
C70 → C68
Trajectory + control
 $x_{ij}(t), u_{ij}(t)$

2 Audit motion

- ✓ Input + speed
- ✓ Mesh clearance
- ✓ Terminal errors

All pass: $x_{ij} = 1$

Fail → omit edge

3 Build graph

SOOA a_{ij}

$C70 \xrightarrow{\ell_{ij}} C68$
Source i View j

Edge: motion + cost

Node: view + mask G_j

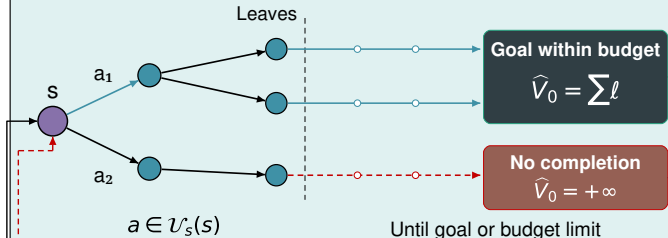
d Rollout ADP

1 Exact lookahead

Safe actions · depth d

2 Full policy tails

Deterministic greedy μ



3 Bellman backup · discard $+\infty$ actions

$$\hat{Q}_d(s, a) = \ell_{ja} + \hat{V}_{d-1}(f(s, a))$$

State update and replan

e Replanning

First action a^*

Minimum finite Q

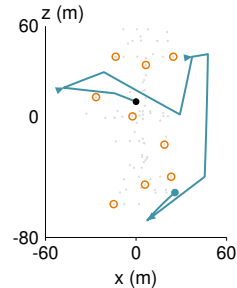
Append arc + update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

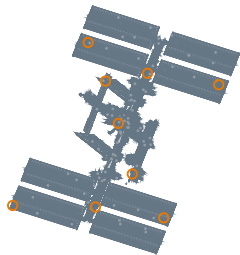
Planned route



Required targets: 9/9

Orbital inspection: sensing, audited transfers and rollout ADP

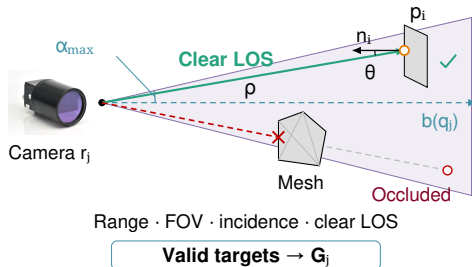
a Mission



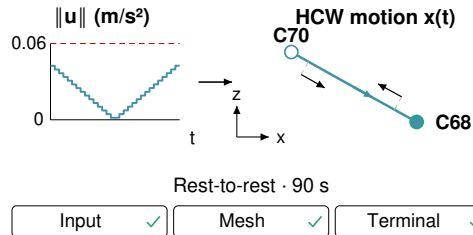
○ Required targets K

Initial state; budget H

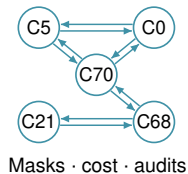
b Camera and LOS



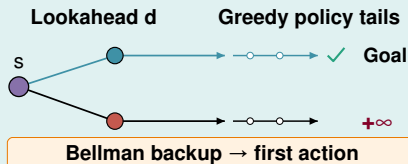
c Controlled HCW transfer



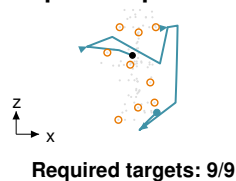
d SOOA graph



e Rollout ADP



f Inspection plan



Rollout ADP with task-aware base-policy evaluation

a Decision state

$$s = (j, m, b, h)$$

j current viewpoint
m covered targets
b selected views
h remaining budget

Audited + unvisited

$$\mathcal{U}_s(s): b_a = 0, \chi_{ja} = 1$$

Safe connecting
actions retained

Fixed required set K

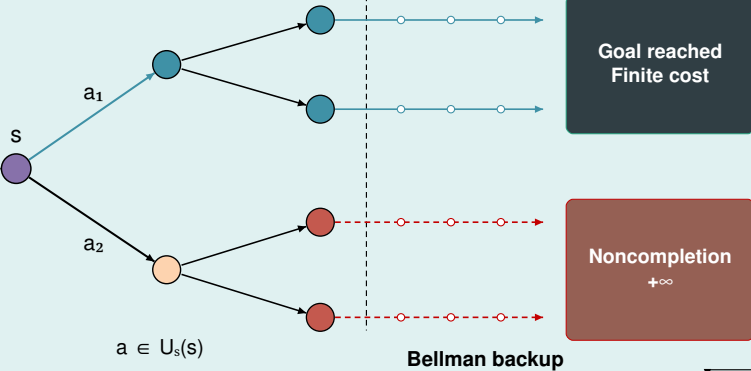
b Rollout ADP

Finite Bellman prefix

Lookahead depth d

Task-aware policy tails

Deterministic greedy μ



$$\hat{Q}_d(s, a) = l_{ja} + \hat{V}_{d-1}(f(s, a))$$

c Policy update

Finite values only

$$a^* = \arg \min_a \hat{Q}_d(s, a)$$

Append first arc and update

$$s' = f(s, a^*)$$

Required goal reached?

Yes No

Complete plan

All $Q = +\infty \rightarrow$ no certified completion

State update and replan